Jedha

CDSD block 2
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Project Steam

Steam's videogames platform

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Goal of the project

Pitch

.**Steam** is a video game digital distribution service and storefront from Valve with software client launched in 2003 to provide game updates automatically for Valve's games, and expanded to distributing third-party titles in late 2005. Steam offers various features, like digital rights management (DRM), game server matchmaking with Valve Anti-Cheat measures, social networking, and game streaming services. Steam client's functions include game update automation, cloud storage for game progress, and community features such as direct messaging, in-game overlay functions and a virtual collectable marketplace. ## Project 🚧

.You're working for Ubisoft, a French video game publisher. They'd like to release a new revolutionary videogame! They asked you conduct a global analysis of the games available on Steam's marketplace in order to better understand the videogames ecosystem and today's trends.

.Goal

.Understand what factors affect the popularity or sales of a video game. And to analyze the video game market globally.

.Adopt different levels of analysis by surveying :

.Analysis at the "macro" level

.Genres analysis

.Platform analysis

.Analysis is performed from a dataset available on aS3 bucket by using Databricks and PySpark to conduct this EDA,
Particularly, you'll have to use Databrick's visualisation tool to create the visualizations.

Load dataset

- Extract tags and dicts
- Flatten
- Feature engineering
- Create special views for analyses

.As the dataset is semi-structured
with a nested schema,
use getField() and explode()

.There are some text
and date fields in this dataset:
utilitary functions are used
💡

.Then agregate functions
and groupBy are used
to conduct segmented analysis.

```
root
 |-- appid: long (nullable = true)
 |-- categories: array (nullable = true)
 |    |-- element: string (containsNull = true)
 |-- ccu: long (nullable = true)
 |-- developer: string (nullable = true)
 |-- discount: string (nullable = true)
 |-- genre: string (nullable = true)
 |-- header_image: string (nullable = true)
 |-- initialprice: string (nullable = true)
 |-- languages: string (nullable = true)
 |-- name: string (nullable = true)
 |-- negative: long (nullable = true)
 |-- owners: string (nullable = true)
 |-- platforms: struct (nullable = true)
 |    |-- linux: boolean (nullable = true)
 |    |-- mac: boolean (nullable = true)
 |    |-- windows: boolean (nullable = true)
 |-- positive: long (nullable = true)
 |-- price: string (nullable = true)
 |-- publisher: string (nullable = true)
```



Analysis at the "macro" level

Which publisher has released the most games on Steam?

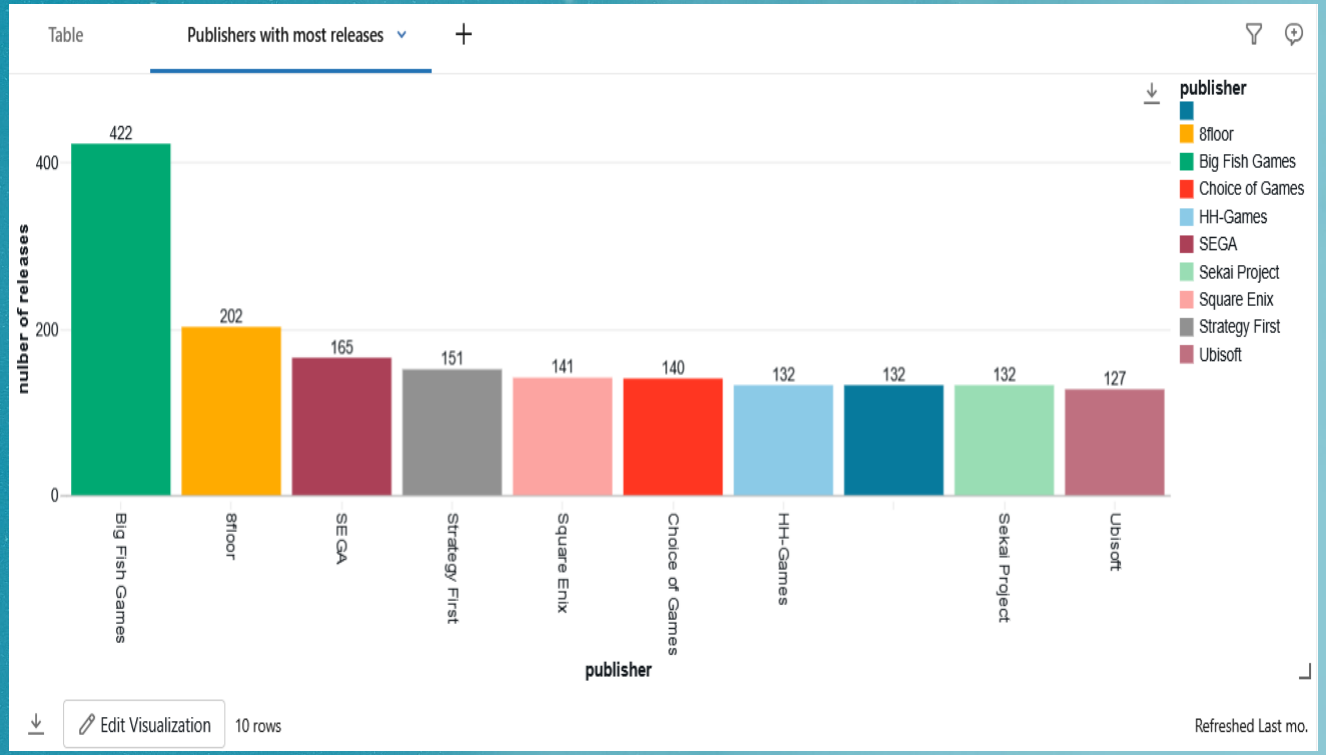
Big Fish games is far beyond more than twice as much the second

8floor with 202 releases

.SEGA

.And a groupes followers around 130 – 150 releases

….



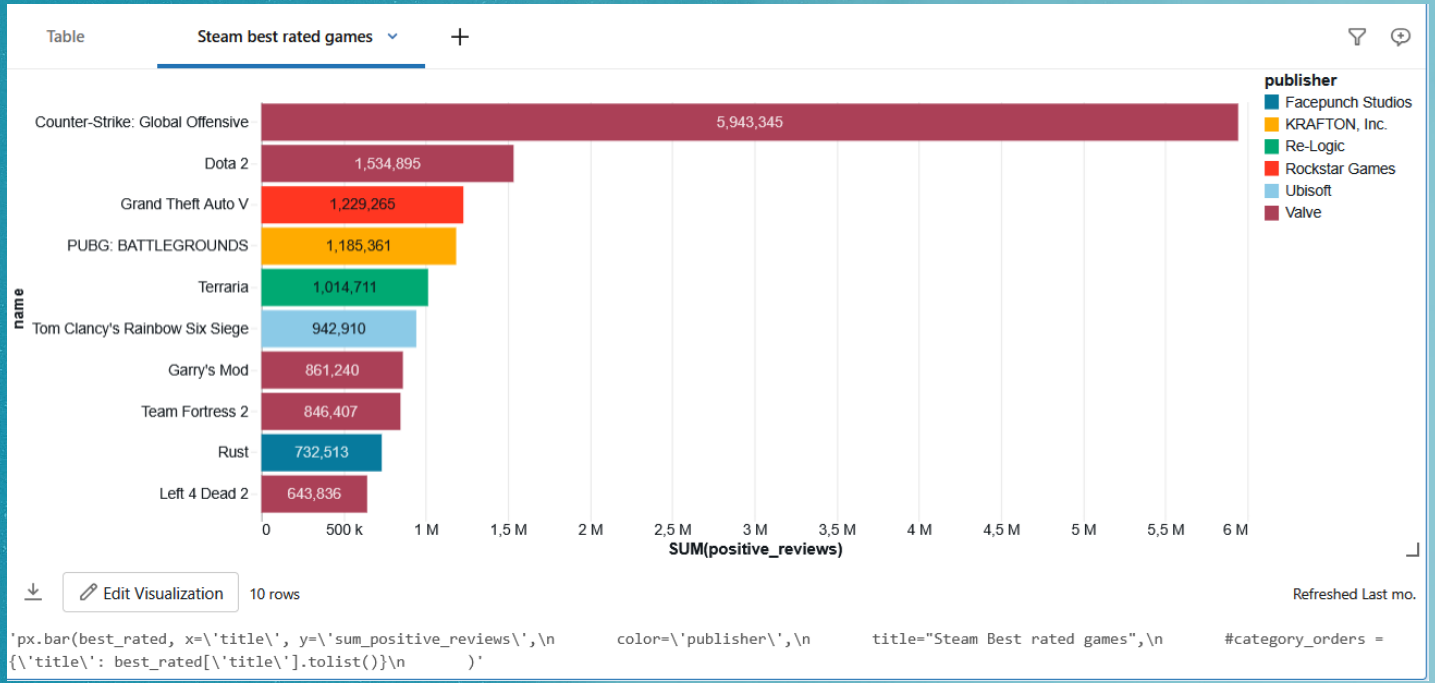
Analysis at the "macro" level

What are the best rated games?

By summing positive reviews

Counter strike global offensive is far beyound the best rated

•



Analysis at the "macro" level

what are the best scored games

Review score ponderated is used :

$$\text{Score Pondéré} = \frac{\sum(X_i \times W_i)}{\sum W_i}$$

Efficient to fight against fraud or clik farms

Best scores are close to 100% !!

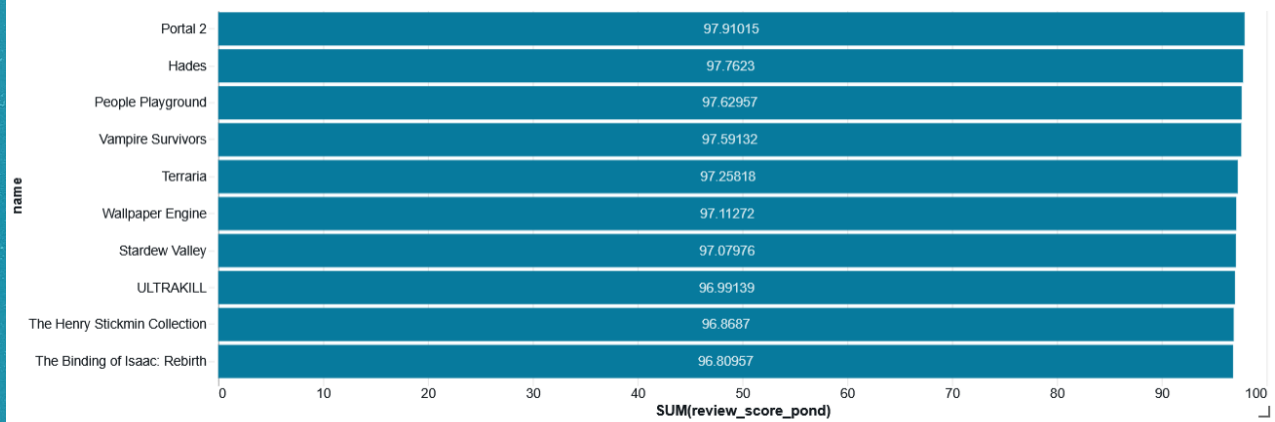
Portal2 is best rated

```
print(f"Top 5 Best ponderated-scored games")
display(best_rated2.limit(10))
```

best_rated2: pyspark.sql.connect.dataframe.DataFrame = [publisher: string, name: string ... 3 more fields]

Top 5 Best ponderated-scored games

Table | Visualization 1 | +



Edit Visualization | 10 rows | Refreshed Last mo.

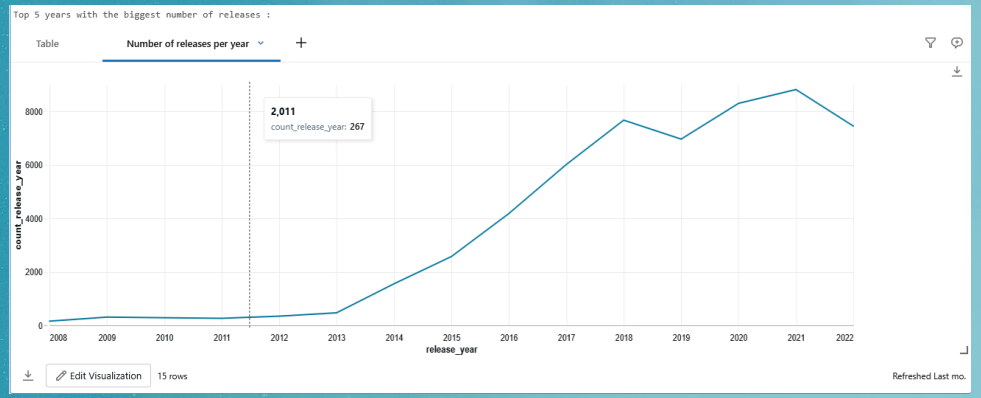
Analysis at the "macro" level

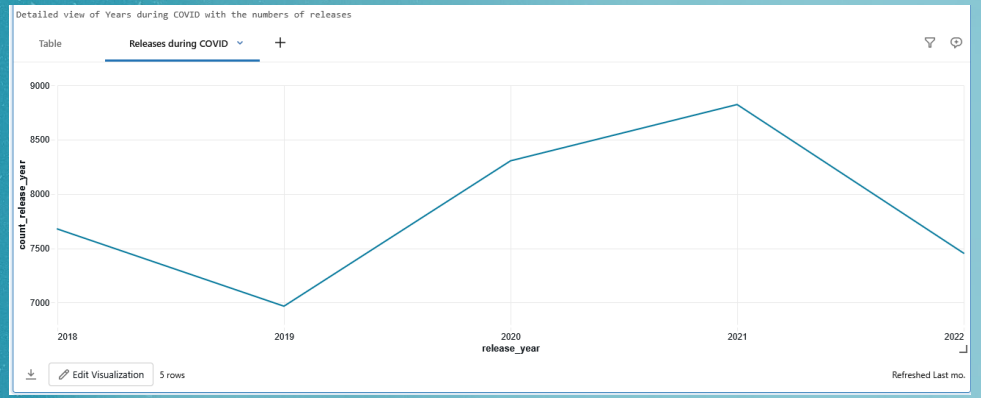
Are there years with more releases?

2021 is a year with high level of releases

Were there more or fewer game releases during the Covid, for example? (between 2020 and 2022)

- No reduction of releases during Covid period,
- even an increase is seen

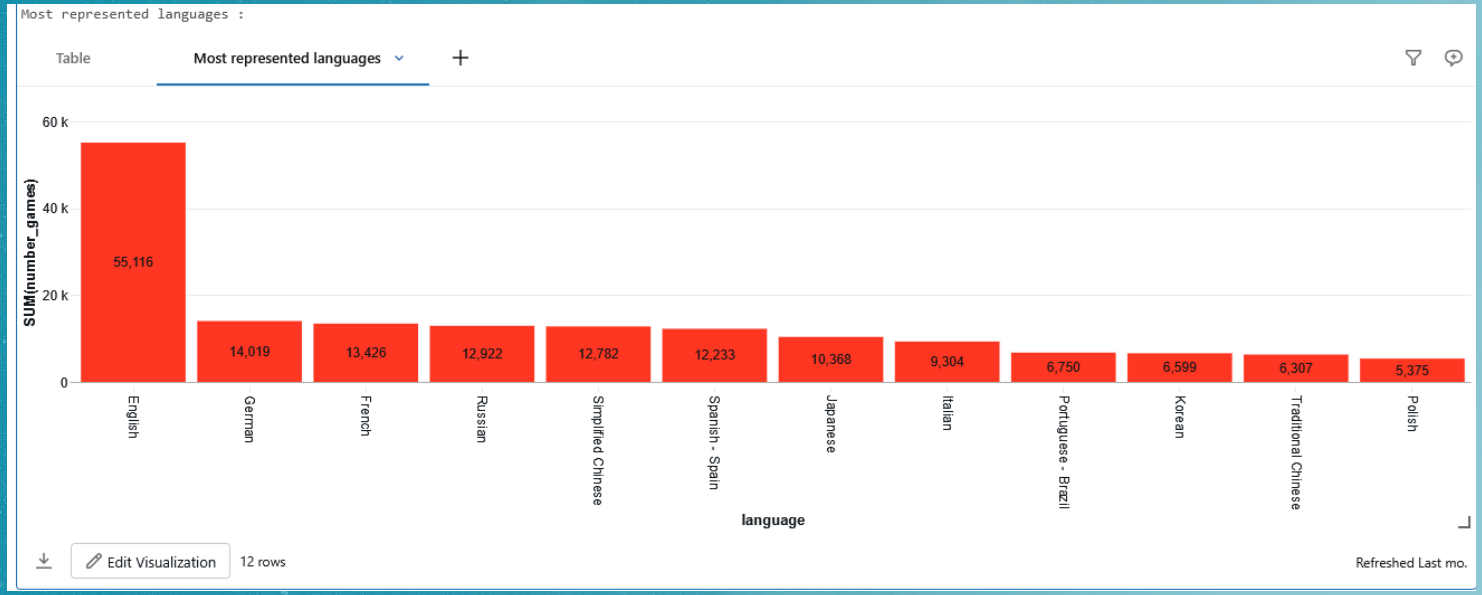




Analysis at the "macro" level

What are the most represented languages?

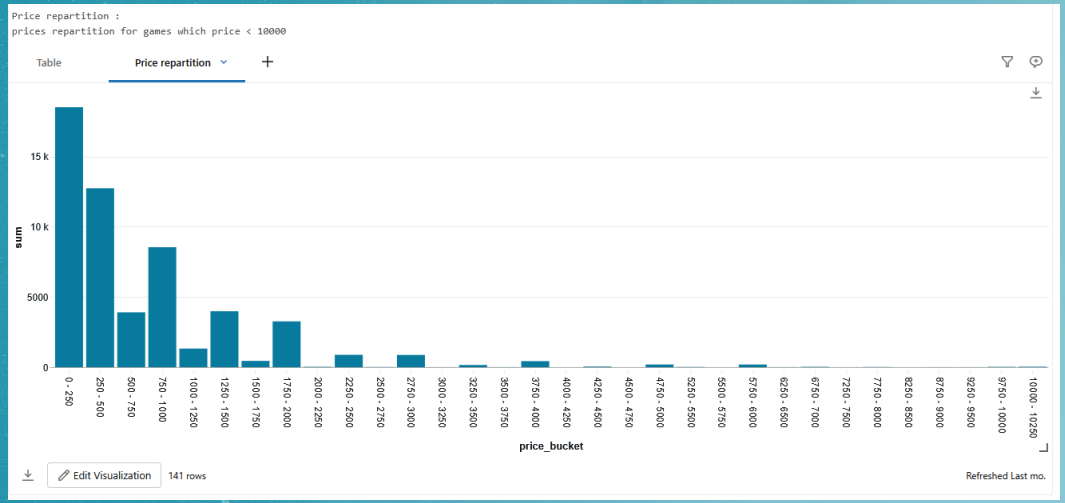
- English at a wide majority
- German
- French
- Russian
-

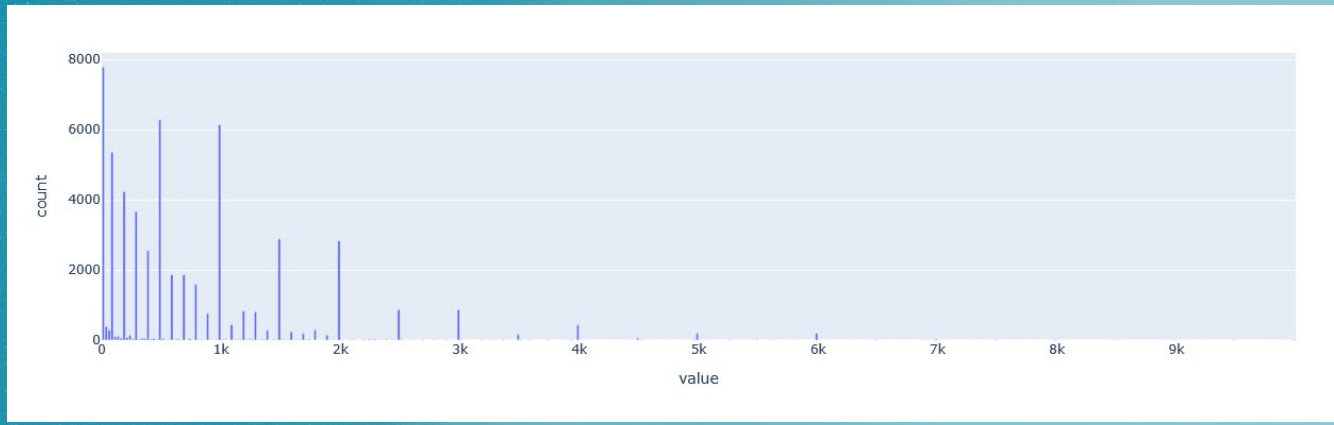


Analysis at the "macro" level

How are the prizes distributed?

- Prices follow a double pattern of 500 and 100
- Best seen on a small price step





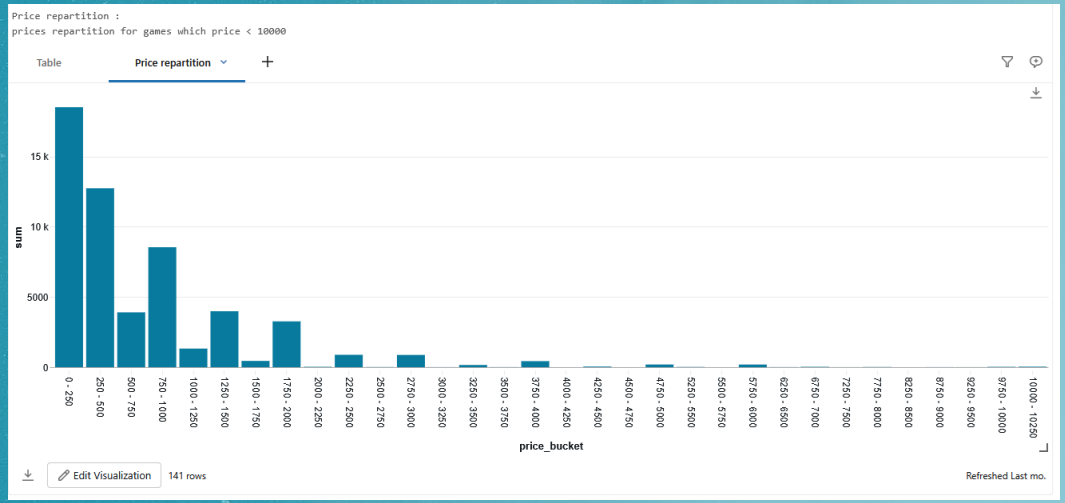
Analysis at the "macro" level

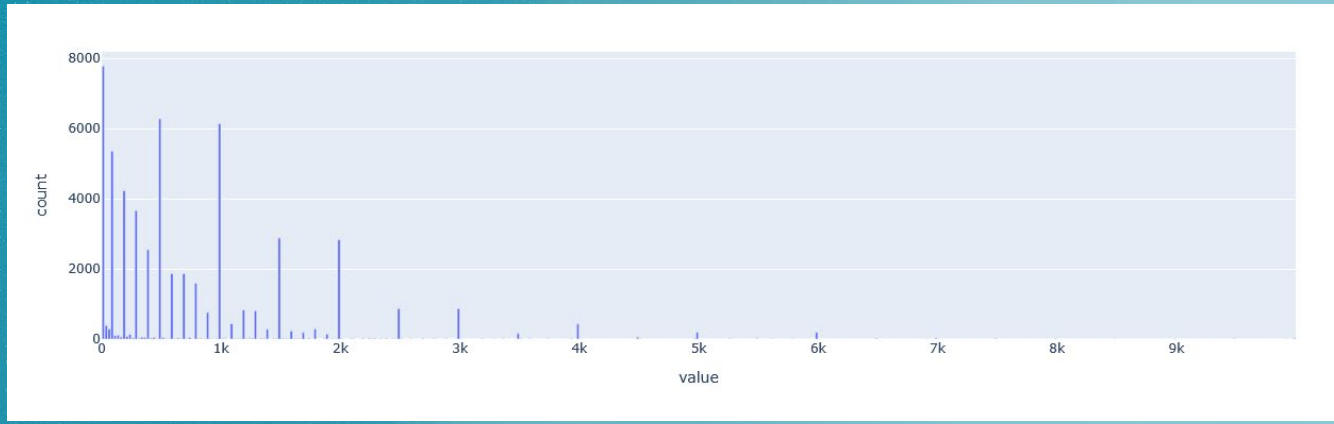
How are the prizes distributed?

- Prices are mainly distributed along 2 families of meshes of 100 and 500

- slowly decreasing

- with a bump around 1000 and 1250

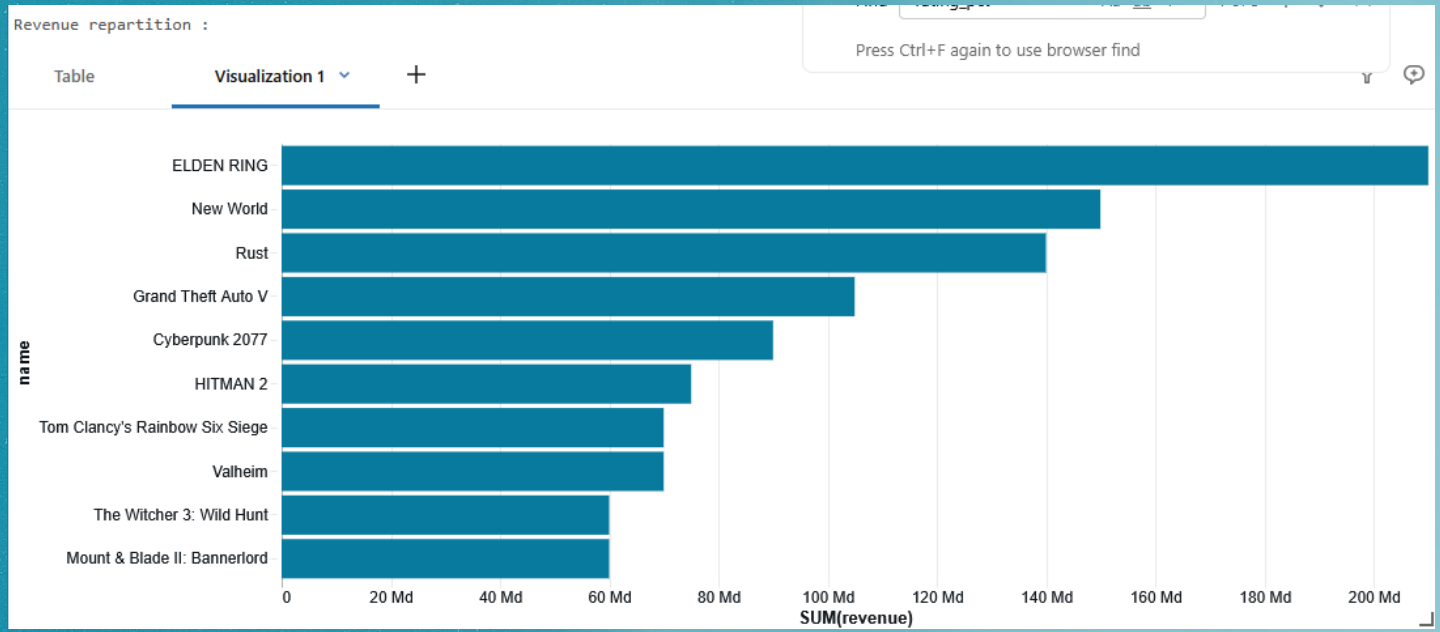




Analysis at the "macro" level

How is revenue distributed

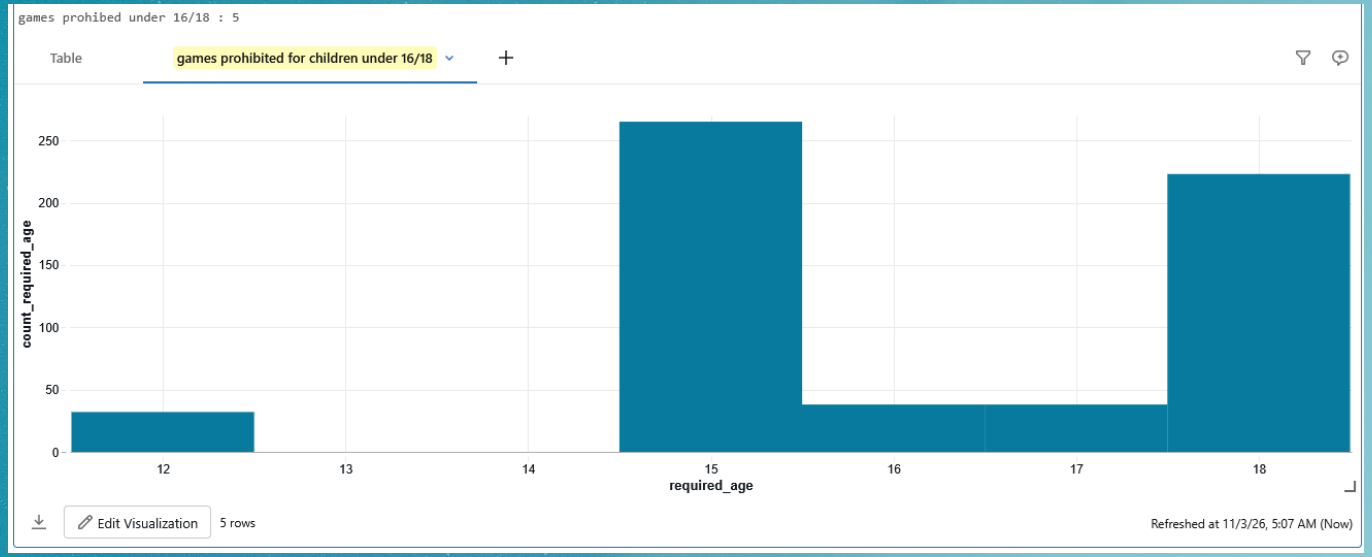
- revenue = price * nb_users_mean
- Revenue is viewed by game name
- Eden ring is the winner !



Analysis at the "macro" level

Are there many games prohibited for children under 16/18?

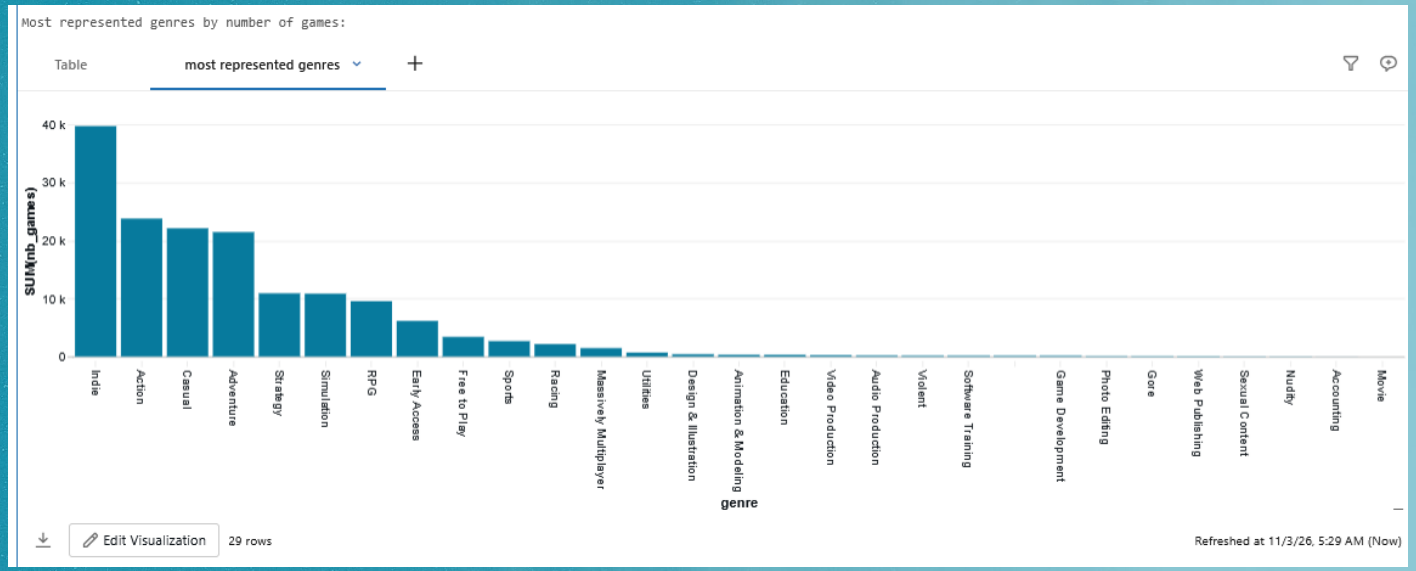
- Main prohibed ages are 16 and 18
- 12 years also are a bit concerned



Genres analysis

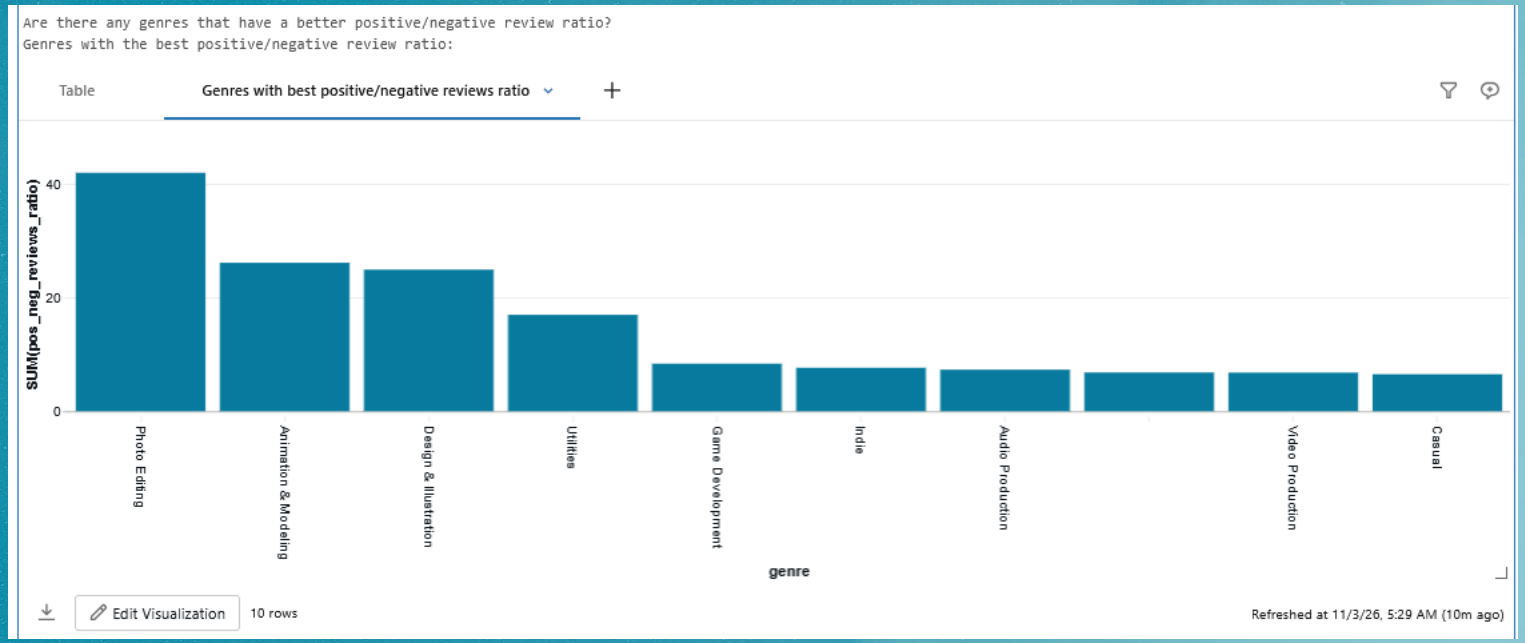
What are the most represented genres?

- Genres are viewed by nb of games
- Most represented genres are Indie,
- then grouped : Action, Casual, Adventure
-



Genres analysis

Are there any genres that have a better positive/negative review ratio?



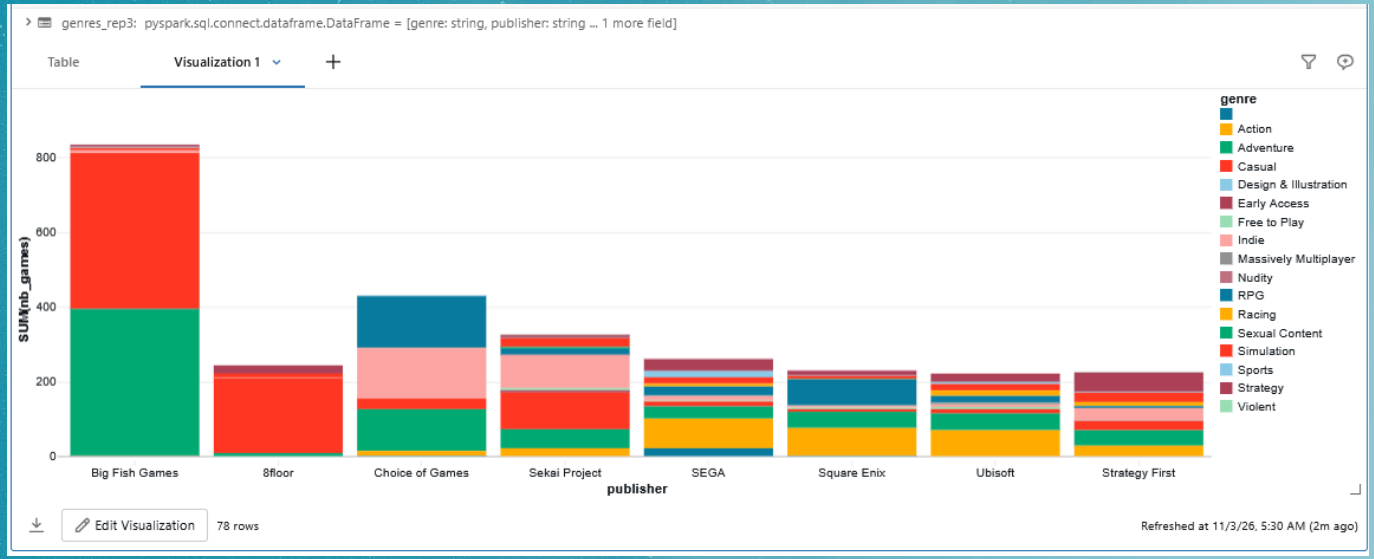
- Best review ratio are for artistic activities, that are not really games :
  - photo editing,
  - animation and modeling,
  - design and illustration

- Genre indie comes only 6th and 10th

Genres analysis

Do some publishers have favorite genres?

- Definitely we can say yes !
- Big Fish prefers adventure and casual
- 8Floor is quasi exclusive casual
- Choice of games is 3/3 : RPG, Indie & adventure

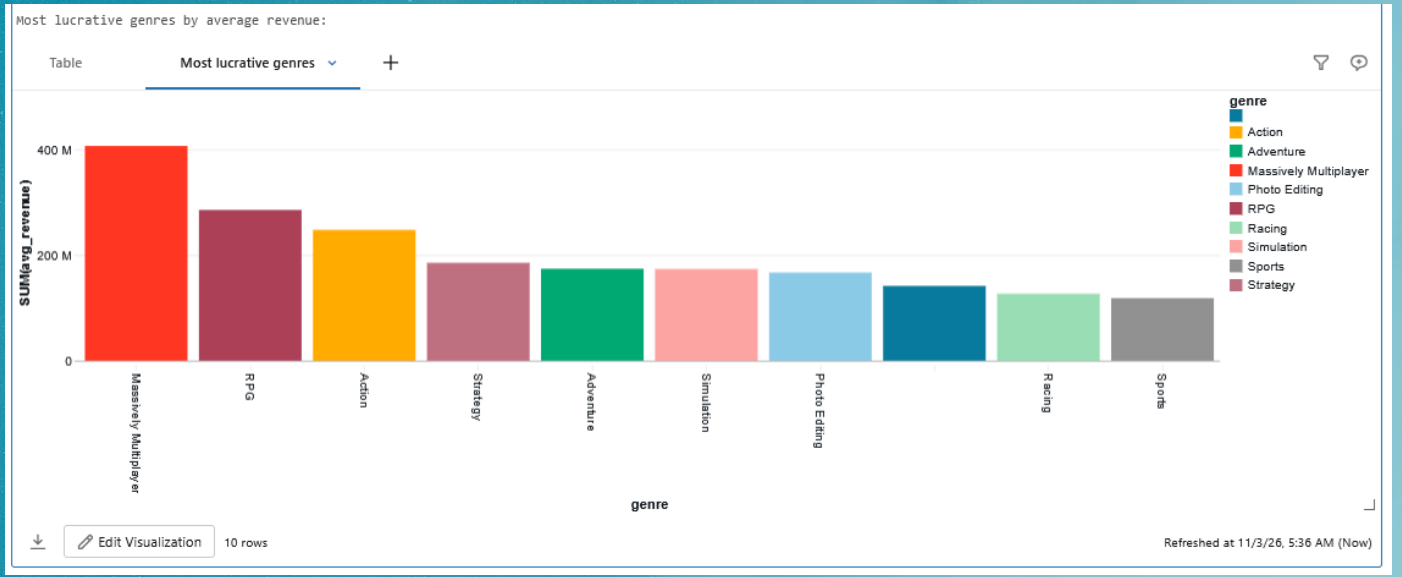


Genres analysis

What are the most lucrative genres?

Most lucrative genres are filtered with average revenue :

- Massive multiplayers
- RPG
- Action
- ...



3 Platform analysis

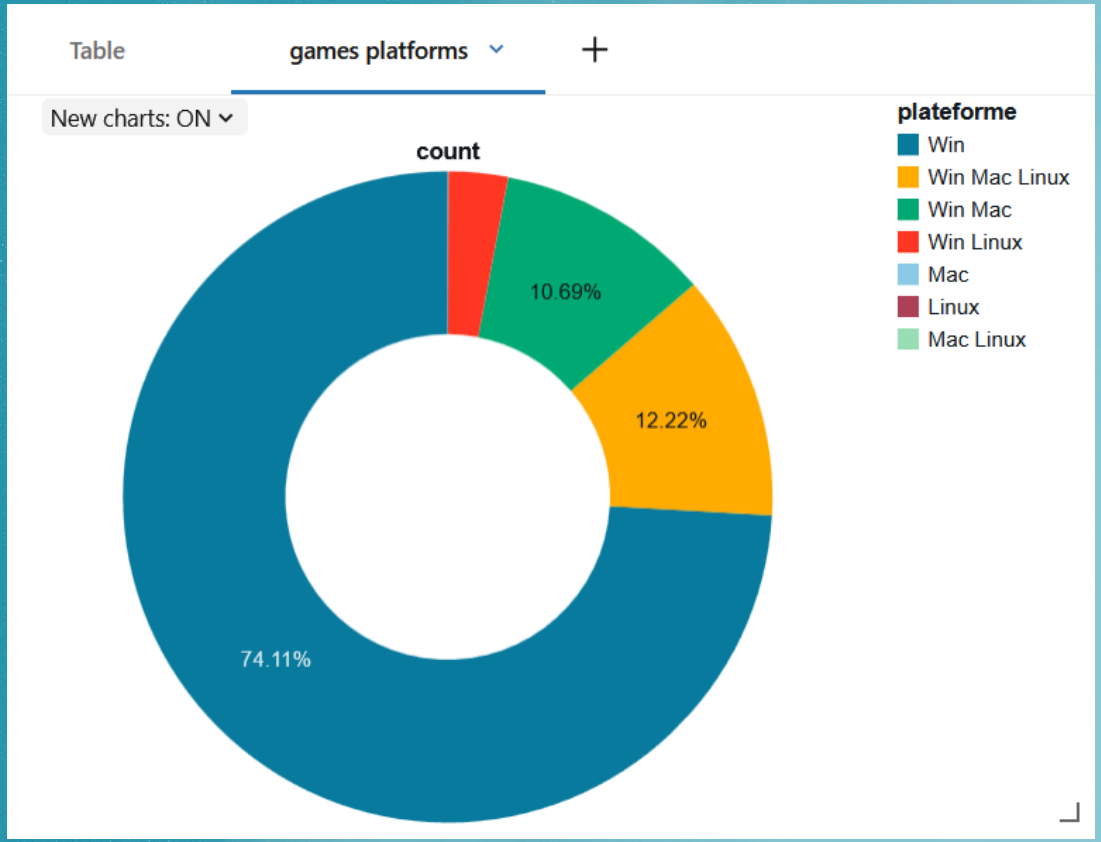
Are most games available on Windows/Mac/Linux instead?

Windows only has a dominating position on platforms ¾ of the offer

Followed by [Win Mac Linux] 12% ; win+Mac 11%

Win + Linux has a very low part below 3%

3 games are linux only !



Platform analysis

Do certain genres tend to be preferentially available on certain platforms?

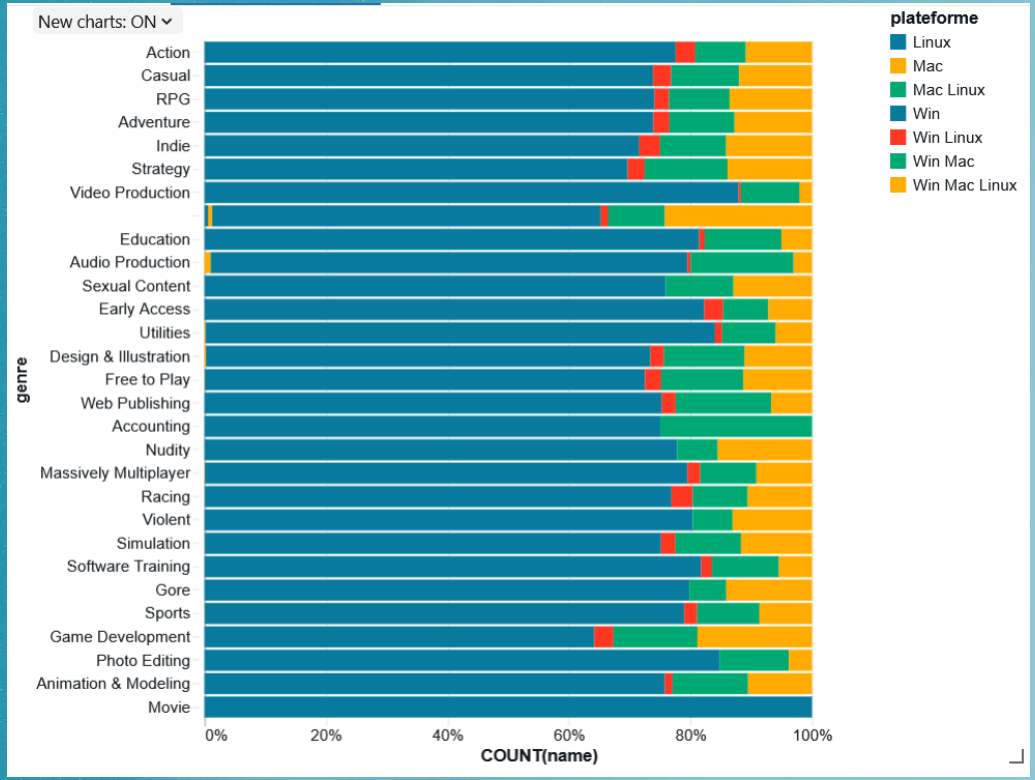
By visualizing genres hiistograms grouped by platform in % one can view all genres, even small volumes on same scale.

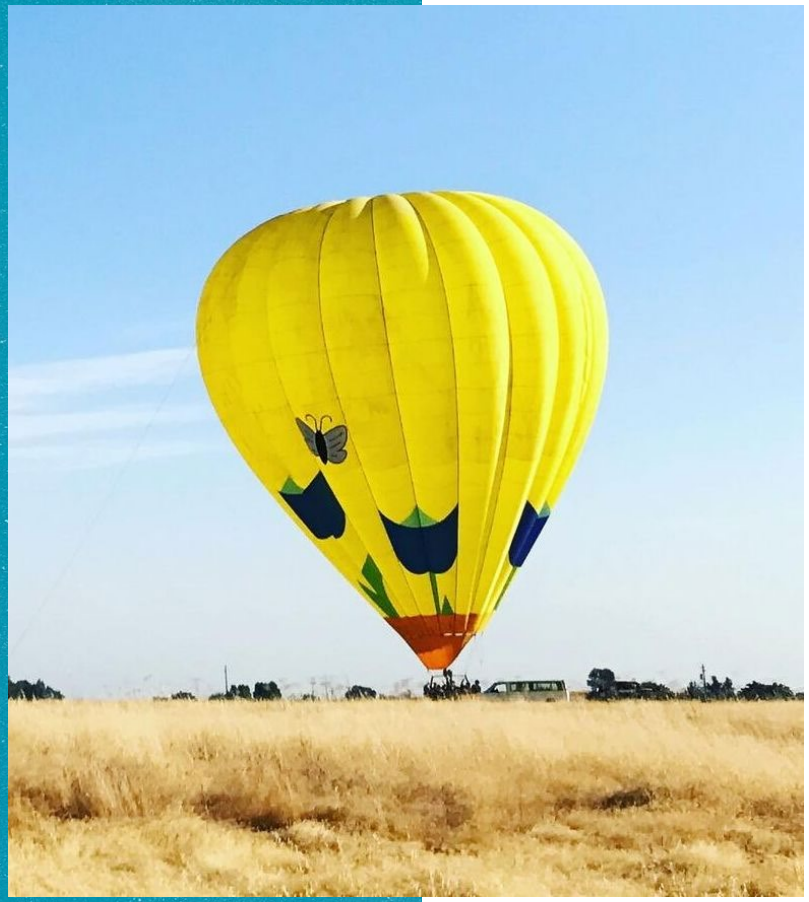
Domination of Windows on almost all genres is visible.

Movies are 100% on Windows,

Accounting has the biggest proportion of Mac

Audio production has a small % of Mac only





Questions ?